Math 110 — Week 6

Solutions

HW6 P1.

Idea. An equation satisfied by P is satisfied by its eigenvalues. For (b), Pv is the range part of v and what is left is in the null space.

1. (a) Let $Pv = \lambda v$, $v \neq 0$. Then $P^2v = P(\lambda v) = \lambda^2 v$. Since $P^2 = P$, $\lambda^2 v = \lambda v$, so $(\lambda^2 - \lambda)v = 0$, so $\lambda(\lambda - 1) = 0$.
2. (b) *Sum.* For any v: $v = (v - Pv) + Pv$. Here $Pv \in \operatorname{range} P$, and $P(v - Pv) = Pv - P^2v = 0$, so $v - Pv \in \operatorname{null} P$. So $V \subseteq \operatorname{null} P + \operatorname{range} P$, and $\operatorname{null} P + \operatorname{range} P \subseteq V$; hence $V = \operatorname{null} P + \operatorname{range} P$.
3. *Direct.* Let $v \in \operatorname{null} P \cap \operatorname{range} P$. Then $v = Pw$ for some w, and $Pv = 0$. So $v = Pw = P^2w = P(Pw) = Pv = 0$. So $\operatorname{null} P \cap \operatorname{range} P = \{0\}$.
4. With the sum step, $V = \operatorname{null} P \oplus \operatorname{range} P$ (Def. 6).

HW6 P2.

Idea. Differentiation lowers the degree of a nonconstant polynomial, and a nonzero multiple of p has the same degree as p.

1. (a) Suppose $p' = \lambda p$ with $p \neq 0$. If $\lambda \neq 0$, then $\deg(\lambda p) = \deg p$, but $\deg p' < \deg p$ (or $p' = 0$ when p is constant, while $\lambda p \neq 0$). Contradiction. So $\lambda = 0$.
2. For $\lambda = 0$: $p' = 0$ means p is constant. So the only eigenvalue is 0, and $E(0, D) = \operatorname{span}(1)$.
3. (b) On smooth functions, $e^{\lambda x}$ satisfies $(e^{\lambda x})' = \lambda e^{\lambda x}$, so every $\lambda \in \mathbb{R}$ is an eigenvalue. The degree argument is not available, and $e^{\lambda x}$ is not a polynomial for $\lambda \neq 0$. (This space is infinite-dimensional, so having infinitely many eigenvalues contradicts nothing.)

HW6 P3.

Idea. $S^{-1}TSv = \lambda v$ becomes $T(Sv) = \lambda(Sv)$ after applying S.

1. Applying S to both sides is reversible because S is invertible, so

$$\begin{aligned} v \in E(\lambda, R) &\iff S^{-1}TSv = \lambda v \iff TSv = \lambda Sv \\ &\iff Sv \in E(\lambda, T) \iff v \in S^{-1}E(\lambda, T). \end{aligned}$$

2. S^{-1} is an isomorphism, so $E(\lambda, R) \neq \{0\}$ exactly when $E(\lambda, T) \neq \{0\}$. So R and T have the same eigenvalues (and the eigenspaces have the same dimensions).

HW6 P4.

Idea. Expand; then eigenvalue $\iff$ $\lambda I - A$ not invertible $\iff$ determinant zero.

1. $tI - A = \left(\begin{smallmatrix} t-a & -b \\ -c & t-d \end{smallmatrix}\right)$, so

$$\det(tI - A) = (t - a)(t - d) - (-b)(-c) = t^2 - (a + d)t + ad - bc.$$

2. λ is an eigenvalue $\iff$ $(A - \lambda I)x = 0$ has a nonzero solution $\iff$ $\lambda I - A$ is not injective $\iff$ $\lambda I - A$ is not invertible (operator on $\mathbb{F}^2$, LADR 5.7) $\iff$ $\det(\lambda I - A) = 0$ (W4 of the handout).

HW6 P5.

Idea. $t^2 - (a+d)t + (ad-bc)$, then one row of $A - \lambda I$.

1. A_1: $\chi = t^2 + 1$. Over $\mathbb{R}$: no roots, no eigenvalues, no eigenvectors. Over $\mathbb{C}$: $\lambda = \pm i$. $A_1(x,y) = (-y, x)$. For $\lambda = i$: $-y = ix$, so $E(i) = \operatorname{span}\big((1,-i)\big)$. For $\lambda = -i$ (conjugate): $E(-i) = \operatorname{span}\big((1,i)\big)$. Two distinct eigenvalues in dimension 2: a basis of eigenvectors exists over $\mathbb{C}$.
2. A_2: $\chi = t^2 - 2t + 1 = (t-1)^2$. Only $\lambda = 1$. $(A_2 - I)(x,y) = (y, 0) = 0$ gives $y = 0$: $E(1) = \operatorname{span}(e_1)$, dimension 1. Every eigenvector is a multiple of e_1, so there is no basis of eigenvectors.
3. A_3: $\chi = t^2 - t - 1$, $\lambda_\pm = \frac{1 \pm \sqrt{5}}{2}$. Eigenvector $(b, \lambda - a) = (1, \lambda)$: check

$$A_3(1,\lambda) = (\lambda, 1+\lambda) = (\lambda, \lambda^2)$$

since $\lambda^2 = \lambda + 1$. $E(\lambda_\pm) = \operatorname{span}\big((1, \lambda_\pm)\big)$. Two distinct real eigenvalues: a basis of eigenvectors exists.

HW6 P6.

Idea. Applying T shifts each $T^k v$ to $T^{k+1} v$. Once one power depends on the earlier ones, all later powers do.

1. (a) An element of $C(v,T)$ is a finite combination $\sum_k a_k T^k v$. Applying T gives $\sum_k a_k T^{k+1} v \in C(v,T)$.
2. (b) Let $n = \dim V$. The $n+1$ vectors $v, Tv, \ldots, T^n v$ are dependent, so there is a smallest m with $T^m v \in \operatorname{span}(v, \ldots, T^{m-1} v) =: U$; $m \geq 1$ because $v \neq 0$, and $m \leq n$.
3. U is T-invariant: T sends $T^k v$ to $T^{k+1} v \in U$ for $k < m-1$, and $T^{m-1} v$ to $T^m v \in U$.
4. So $T^k v \in U$ for all k (induction: $T^{k+1} v = T(T^k v) \in T(U) \subseteq U$). Hence $C(v,T) \subseteq U$, and $U \subseteq C(v,T)$ is clear from the definition.

HW6 P7.

Idea. Solve $(P - I)p = 0$ and normalise. Here the rows also sum to 1, so the uniform vector works.

1. $Pp = p$: $\frac{1}{2}p_1 + \frac{1}{2}p_2 = p_1$ gives $p_2 = p_1$; $\frac{1}{2}p_2 + \frac{1}{2}p_3 = p_3$ gives $p_3 = p_2$.
2. With $p_1 + p_2 + p_3 = 1$: $p = (\frac{1}{3}, \frac{1}{3}, \frac{1}{3})$. Check the middle row: $\frac{1}{2} \cdot \frac{1}{3} + \frac{1}{2} \cdot \frac{1}{3} = \frac{1}{3}$.
3. Meaning: if the chain starts with each state equally likely, then at every later time each state is still equally likely, since $P^k p = p$ for all k.

R1.

Idea. Powers first, then combine.

1. $T^k v = \lambda^k v$ for all $k \geq 0$: true for $k = 0$, and

$$T^{k+1} v = T(\lambda^k v) = \lambda^k T v = \lambda^{k+1} v.$$

2. For $p(z) = \sum a_k z^k$:

$$p(T)v = \sum a_k T^k v = \sum a_k \lambda^k v = p(\lambda) v.$$

R2.

Idea. Use $(T'\varphi)(u) = \varphi(Tu)$.

1. If $u \in \operatorname{null} T$ then $Tu = 0 \in \operatorname{null} T$. If $u \in \operatorname{range} T$ then $Tu \in \operatorname{range} T$, being an output of T.
2. Let $\varphi \in (\operatorname{null} T)^0$. For $u \in \operatorname{null} T$: $(T'\varphi)(u) = \varphi(Tu) = \varphi(0) = 0$. So $T'\varphi \in (\operatorname{null} T)^0$.
3. Let $\varphi \in (\operatorname{range} T)^0$. For any v: $(T'\varphi)(v) = \varphi(Tv) = 0$ because $Tv \in \operatorname{range} T$. So $T'\varphi = 0$, which lies in $(\operatorname{range} T)^0$.

R3.

Idea. (a) and (b) fail already in $\mathbb{R}^2$: (a) for a diagonal matrix with two different entries, (b) for the rotation by 90°. For (c), the identity has one eigenspace, the whole space.

1. (a) False. For $\operatorname{diag}(1,2)$, e_1 and e_2 are eigenvectors, and $e_1+e_2 \mapsto (1,2)$, which is not a multiple of $(1,1)$. (Also v_1+v_2 could be 0, for $v_2=-v_1$.)
2. (b) False. Rotation by 90° on $\mathbb{R}^2$ has no eigenvector (Ex. 49).
3. (c) True. For $T=I$, $E(1,I)=V$, so V is the direct sum of the single eigenspace $E(1,I)$.

R4.

Idea. Finite dimension gives a relation among $v, Tv, \dots, T^n v$, that is, a polynomial with $p(T)v=0$. The Fundamental Theorem factors p; the eigenvector is $q(T)v$, not v.

1. (a) Every nonconstant polynomial with *complex* coefficients has a zero in $\mathbb{C}$.
2. (b) Let $n = \dim V$. The $n+1$ vectors $v, Tv, \dots, T^n v$ are dependent, so some nontrivial relation $a_0 v + a_1 Tv + \cdots + a_m T^m v = 0$ with $a_m \neq 0$ holds; that is, $p(T)v=0$ for $p(z) = a_0 + \cdots + a_m z^m$. If $m=0$, the relation is $a_0 v = 0$ with $a_0 \neq 0$, so $v=0$, which is excluded. So $m \geq 1$ and p is nonconstant.
3. (c) Choose such a relation with m smallest. By (a), p has a zero λ, so $p(z) = (z-\lambda)q(z)$ with $\deg q = m-1$ and leading coefficient $a_m \neq 0$.
4. Let $w = q(T)v$. If $w=0$, then $q(T)v=0$ is a nontrivial relation of degree $m-1<m$, contradicting the choice of m. So $w \neq 0$.
5. $(T-\lambda I)w = (T-\lambda I)q(T)v = p(T)v = 0$, so $Tw = \lambda w$: w is an eigenvector. (v itself need not be one: for $T=\operatorname{diag}(1,2)$ and $v=(1,1)$, $p=(z-1)(z-2)$ has $p(T)v=0$, but $Tv=(1,2)$.)

R5.

Idea. (a) Each basis function is killed by one simple polynomial in d, so their product kills all of V; an eigenvalue must be a real root of that product. (b) The 2×2 formula.

1. (a) $d(1)=0$, $(d^2+1)\sin x = (d^2+1)\cos x = 0$ and $(d^2+4)\sin 2x = (d^2+4)\cos 2x = 0$, since $(\sin kx)'' = -k^2 \sin kx$ and $(\cos kx)'' = -k^2 \cos kx$.
2. Let $p(z) = z(z^2+1)(z^2+4)$. Polynomials in d commute (LADR 5.17), so $p(d)$ is each of the three factors applied last; hence $p(d)$ kills each of the five spanning functions, and so $p(d)=0$ on V.
3. If $df = \lambda f$ with $f \neq 0$ and $\lambda \in \mathbb{R}$, then $0 = p(d)f = p(\lambda)f$, so $p(\lambda) = \lambda(\lambda^2+1)(\lambda^2+4) = 0$. The last two factors are positive, so $\lambda = 0$.
4. $\lambda=0$ is an eigenvalue: $d(1)=0$. $E(0,d) = \{f \in V : f' = 0\}$ is the constants: $E(0,d) = \operatorname{span}(1)$.
5. (b) In the standard basis T has the matrix $\left[\begin{smallmatrix} 0 & 1 \\ -1 & 0 \end{smallmatrix}\right]$, so $\chi = t^2+1$ and $\lambda = \pm i$.
6. For $\lambda = i$ the recipe $(b, \lambda - a)$ gives $(1,i)$; check $T(1,i) = (i,-1) = i(1,i)$. For $\lambda = -i$ it gives $(1,-i)$; check $T(1,-i) = (-i,-1) = -i(1,-i)$. So $E(i) = \operatorname{span}((1,i))$ and $E(-i) = \operatorname{span}((1,-i))$.

R6.

Idea. Move S past T using $ST = TS$.

1. $\operatorname{null} T$: if $Tv=0$, then $T(Sv) = S(Tv) = S0 = 0$, so $Sv \in \operatorname{null} T$.
2. $\operatorname{range} T$: if $w = Tu$, then $Sw = STu = T(Su) \in \operatorname{range} T$.
3. $E(\lambda, T)$: if $Tv = \lambda v$, then $T(Sv) = S(Tv) = S(\lambda v) = \lambda Sv$, so $Sv \in E(\lambda, T)$.

E1.

Idea. r is the unique solution of a 3×3 linear system; the second equation gives r_2 in terms of r_1, and the third then gives r_3 in terms of r_1, which reduces everything to one unknown.

1. The link matrix and damping parameter are

$$P = \begin{bmatrix} 0 & 0 & 1 \\ 1/2 & 0 & 0 \\ 1/2 & 1 & 0 \end{bmatrix}, \qquad \alpha = \tfrac{1}{2},$$

   so $(1-\alpha)v = (\frac{1}{6}, \frac{1}{6}, \frac{1}{6})$.

2. The three coordinate equations of $r = \frac{1}{2}Pr + \frac{1}{6}\mathbf{1}$:

$$\begin{aligned} r_1 &= \tfrac{1}{2}r_3 + \tfrac{1}{6}, \\ r_2 &= \tfrac{1}{4}r_1 + \tfrac{1}{6}, \\ r_3 &= \tfrac{1}{4}r_1 + \tfrac{1}{2}r_2 + \tfrac{1}{6}. \end{aligned}$$

3. Substitute r_2 into the third:

$$r_3 = \tfrac{1}{4}r_1 + \tfrac{1}{8}r_1 + \tfrac{1}{12} + \tfrac{1}{6} = \tfrac{3}{8}r_1 + \tfrac{1}{4}.$$

4. Substitute into the first:

$$r_1 = \tfrac{1}{2}\left(\tfrac{3}{8}r_1 + \tfrac{1}{4}\right) + \tfrac{1}{6} = \tfrac{3}{16}r_1 + \tfrac{7}{24}, \qquad \text{so} \qquad \tfrac{13}{16}r_1 = \tfrac{7}{24}, \quad r_1 = \tfrac{14}{39}.$$

5. The other coordinates:

$$r_2 = \tfrac{1}{4} \cdot \tfrac{14}{39} + \tfrac{1}{6} = \tfrac{14}{156} + \tfrac{26}{156} = \tfrac{10}{39}, \qquad r_3 = \tfrac{3}{8} \cdot \tfrac{14}{39} + \tfrac{1}{4} = \tfrac{42}{312} + \tfrac{78}{312} = \tfrac{15}{39}.$$

6. So $r = \frac{1}{39}(14, 10, 15)$; the coordinates sum to 1. Ranking: page 3, then 1, then 2.